

UCLA

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GoFish!: A Tension Regulation Fishing Robot

TEAM 17

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Overview

Project Overview & Motivation

- **Project Goal**

- Develop a 1-DOF rigid robotic fishing rod
- Simulate fish-fighting interaction under controlled testing
- Develop a control paradigm to regulate fishing-line tension near a 3.5 N target with a ± 0.1 N deadband.

- **Motivation / Literature**

- Prior fishing haptic devices show that force feedback improves realism in fishing simulation.
- Fishing-rod haptic displays have also been studied for presenting force sensation through a rod-and-line structure.

- **Applications**

- Fishing training simulator
- Entertainment / interactive game controller
- Robotics force-feedback education

- **System Concept**

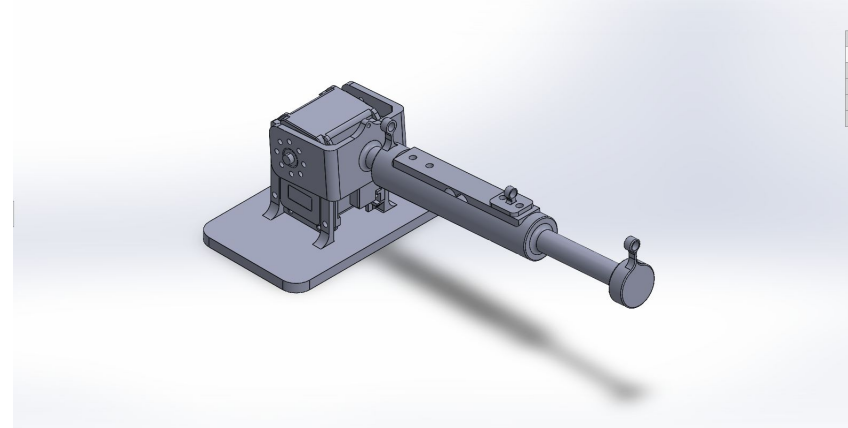
$$F_{pull} \rightarrow T_f \rightarrow \theta_{cmd} \rightarrow \theta$$

Hardware

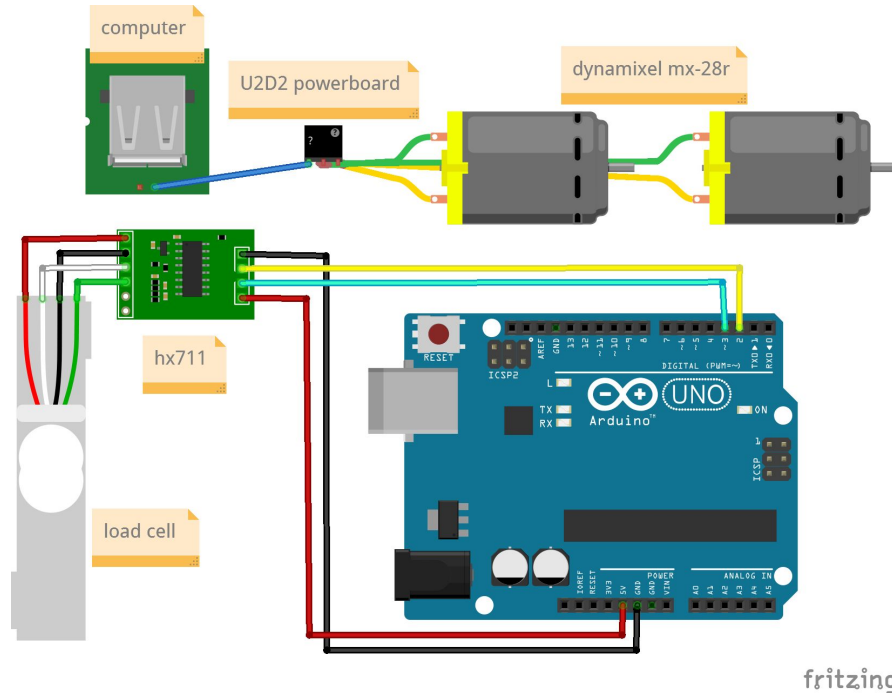
Hardware Test



GoFish: Tension Regulation Fishing Robot

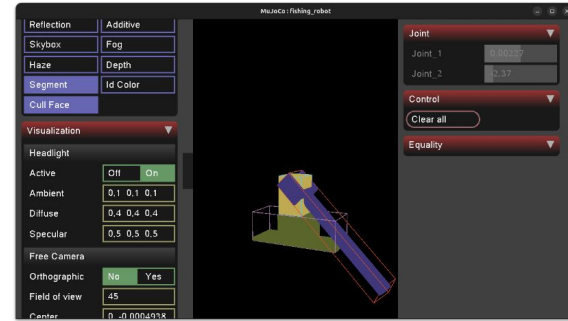
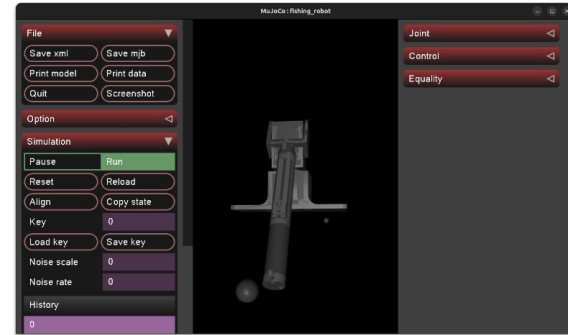
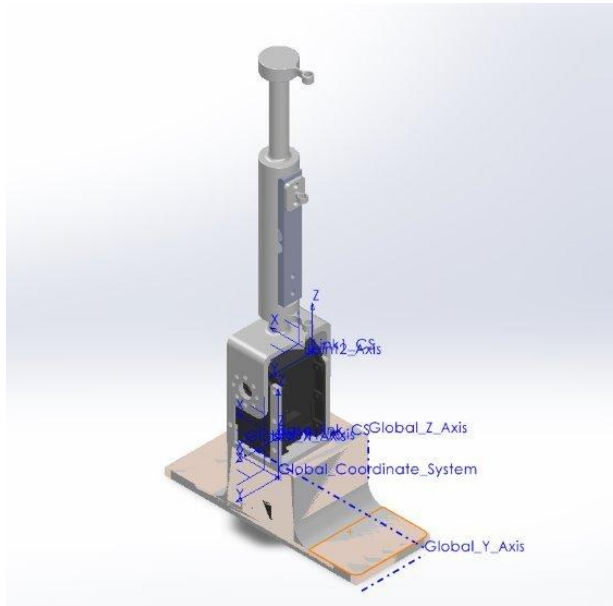


Wiring Diagram



Software

Simulation (mujoco)



Codebase (true implementation)

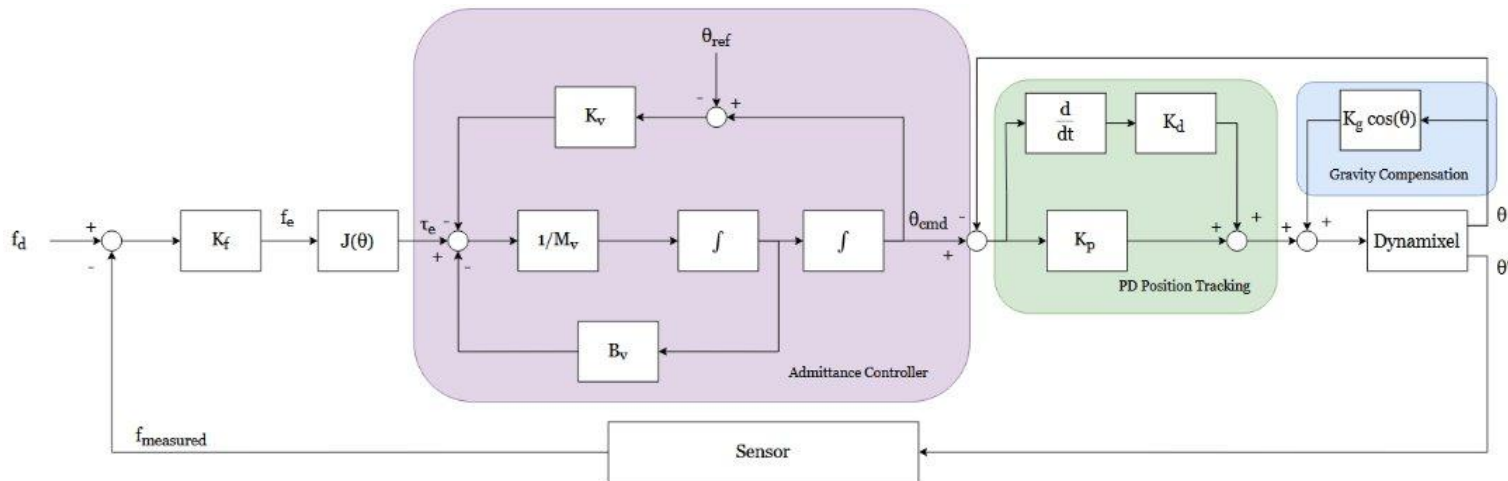
ROS2 - software architecture modularity

Building out the ROS2 codebase such that the control logic interacts seamlessly with Mujoco simulation and hardware control

- *Bringup*: launch configuration. Can launch simulation or hardware control based on preference
- *Description*: mesh and model descriptors
- *Control*: tension control package
- *Planning*: will likely be used to control randomness of 'fish' behavior. Can be thought of as a fish agent (our disturbance)
- *Sensors*: sensor readings and publishing
- *Interfaces*: develop protocol for inter-node communication

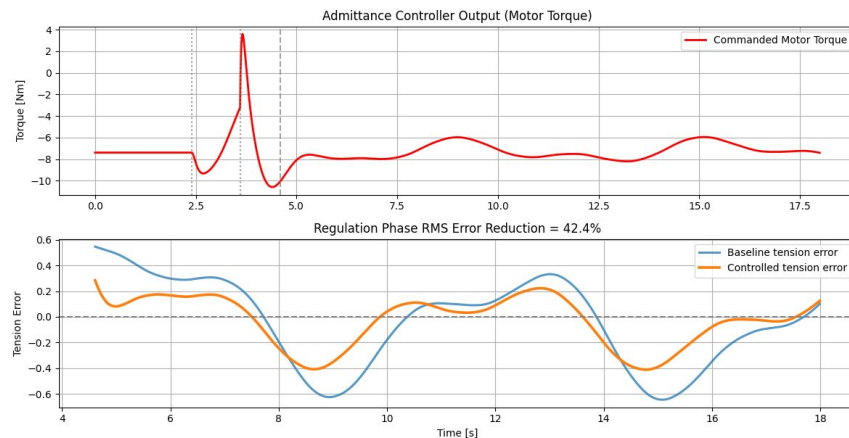
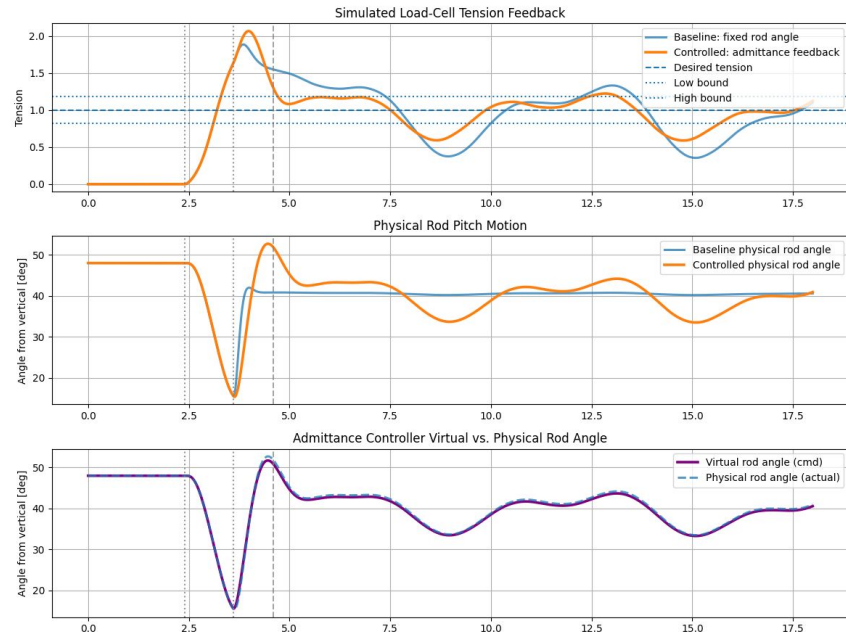
Controller & Simulation

Controller Diagram (Admittance Control)

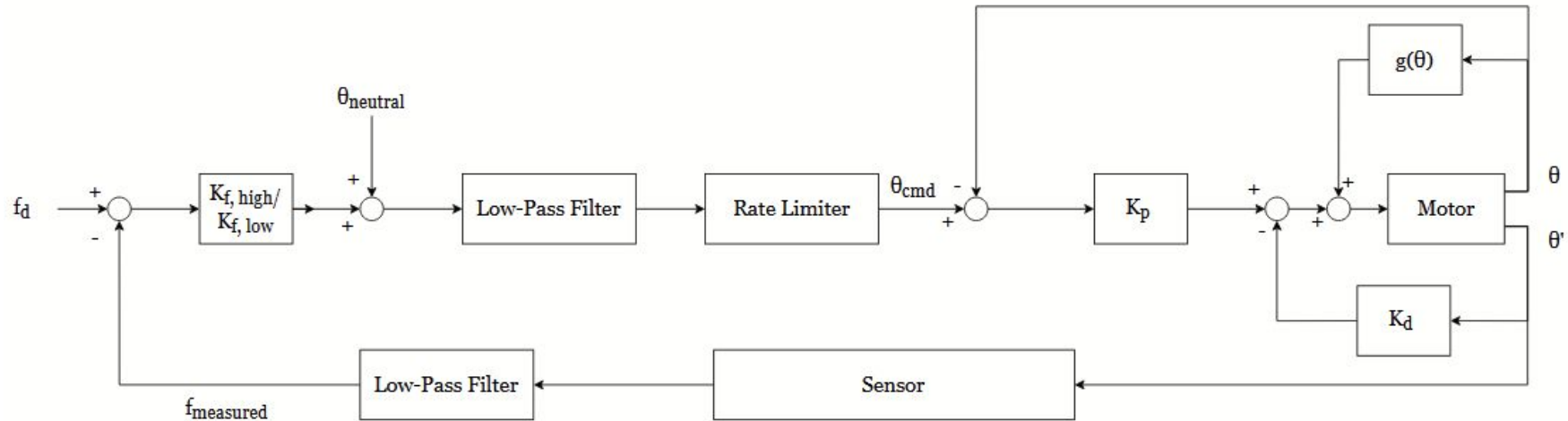


$$\mathbf{M}_v \ddot{\theta}_{cmd} + \mathbf{B}_v \dot{\theta}_{cmd} + \mathbf{K}_v (\theta_{cmd} - \theta_{ref}) = -\mathbf{J}(\theta) \mathbf{K}_f (f_d - f_{measured})$$

Admittance Controller Results

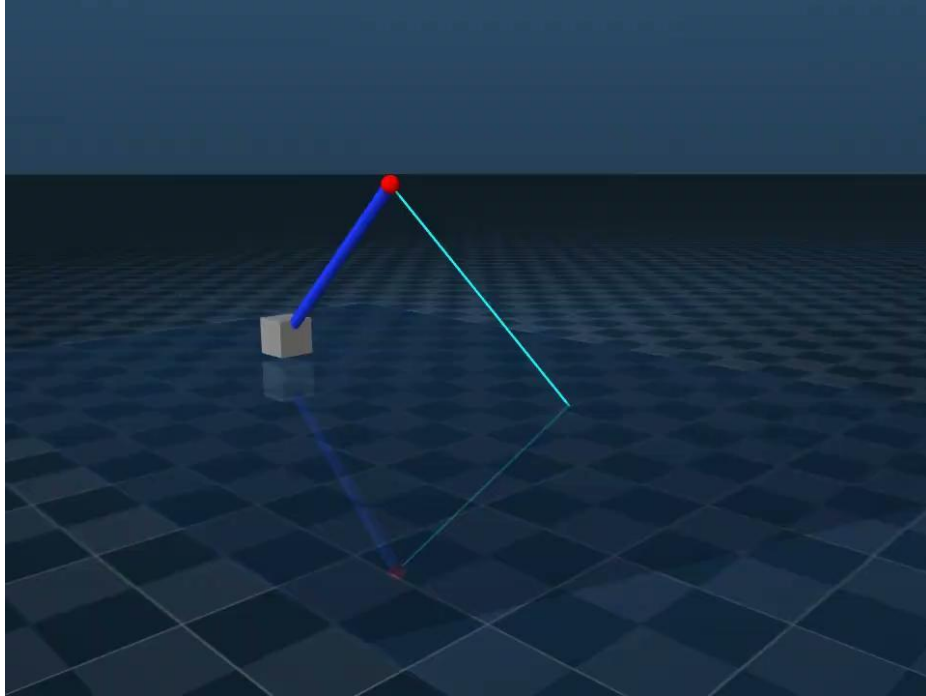


Controller Diagram (Proportional Force Control)

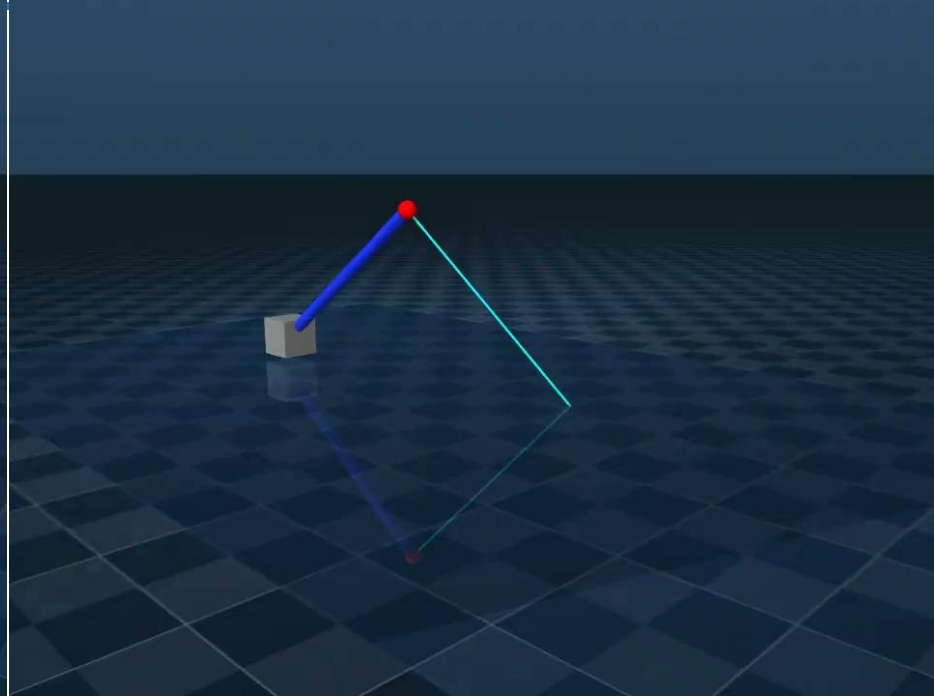


$$K_{f, \text{low/high}}(f_d - f_{\text{measured}}) + \theta_{\text{neutral}} = \theta_{\text{cmd}}$$

Admittance Controller

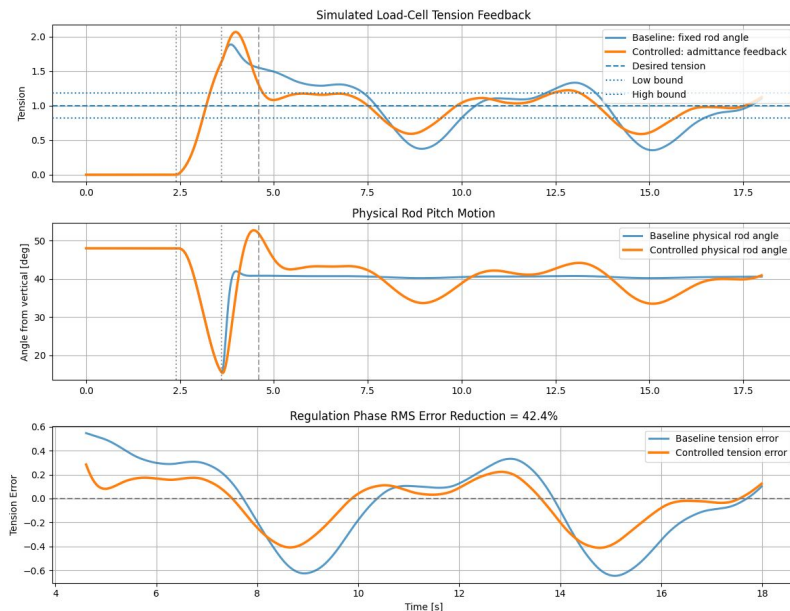


Force Controller

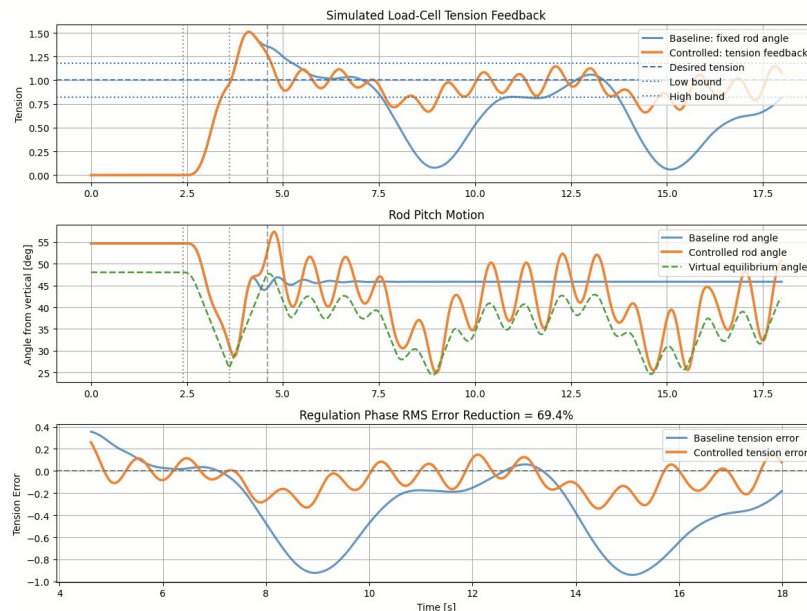


Controller Comparison

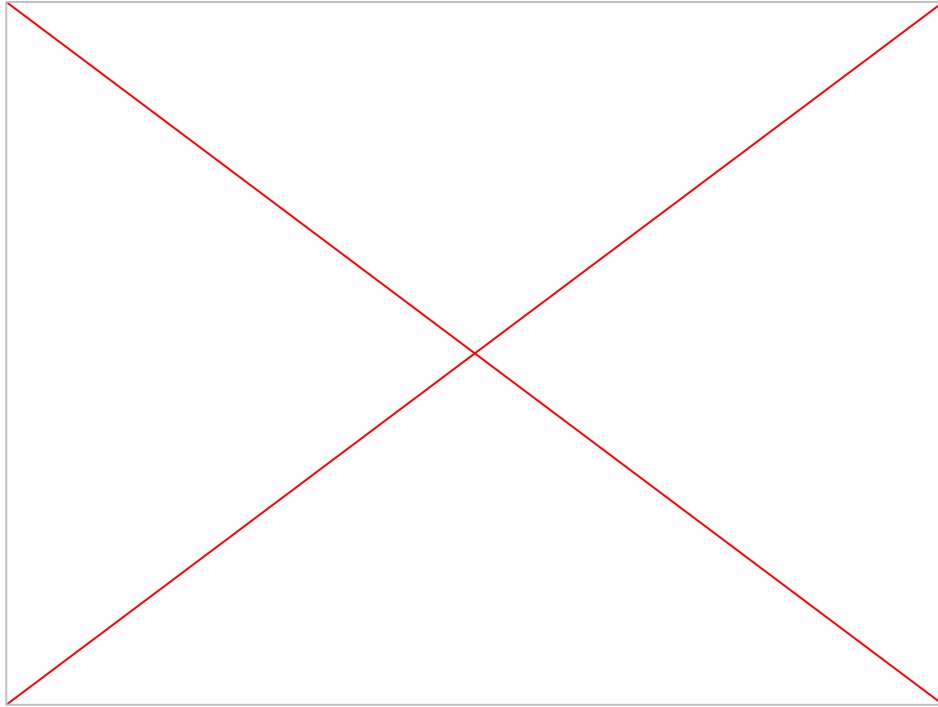
Admittance Control



Proportional Force Control



Results

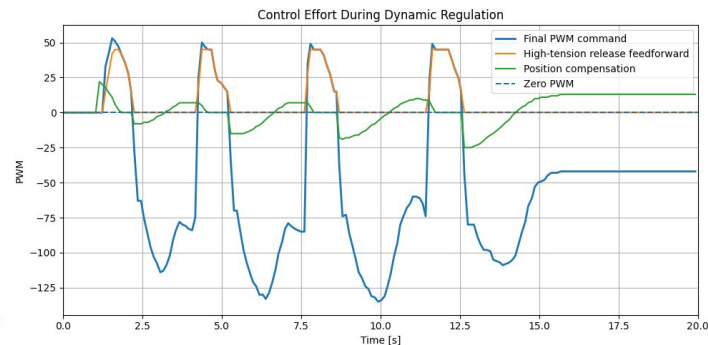
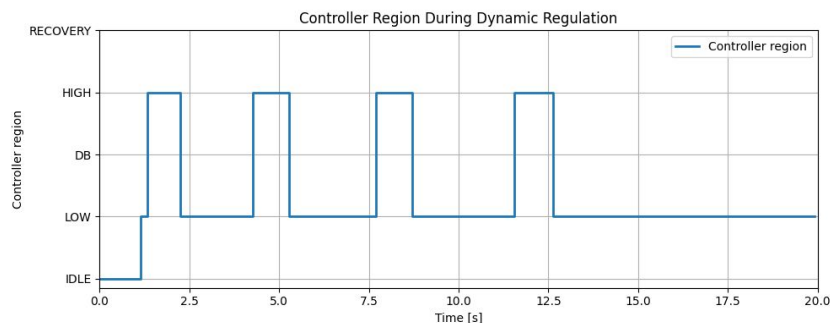


Manual pulling creates line-tension changes, and the rod responds through the Dynamixel motor.

Dynamic Controller Response

Test condition: Rapid manual pulls were applied to create sudden line-tension changes.

Key result: the controller switched between release and pull-back actions, and the PWM command changed accordingly.



Post-Transient Regulation Accuracy

Test condition:

Initial manual pull followed by steady holding to evaluate whether the controller can return the tension near the 3.5 N target.

Target tension: 3.5 N

Final 5 s window:

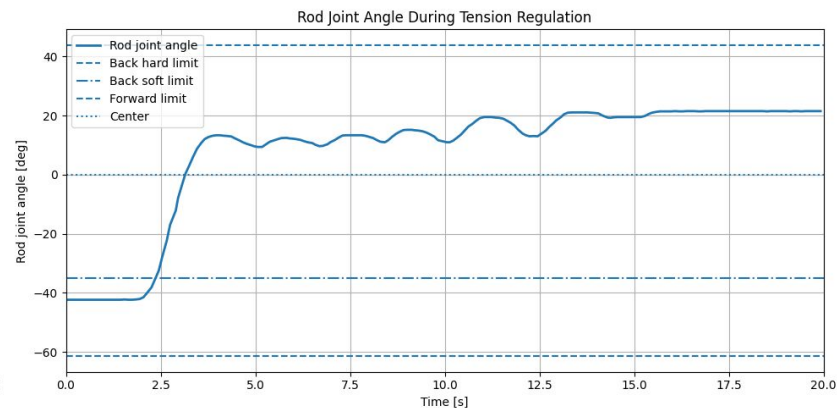
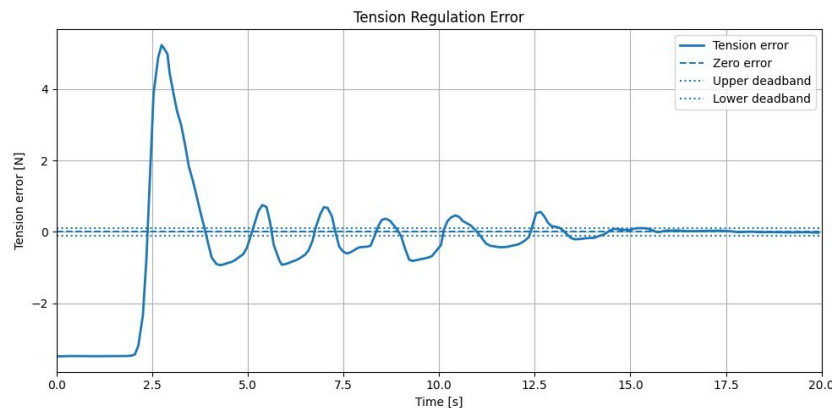
Mean tension: 3.52 N

MAE: 0.025 N (0.72%)

RMSE: 0.036 N (1.03%)

Takeaway:

After the initial engagement and correction phase, the controller reduced the tension error and maintained the measured tension close to the target



Future Developments

In truth, the fishing problem is extremely control rich
in my opinion
I will try to convey this nuance

Algorithmic extensions

2D Tension control

Planarized fish movement adds sensing complexity

- Our load sensor is 1D
 - instead of direct linear force/tension estimation relation, need to be aware of tension vector
 - use load sensor info and kinematic state (encoders) -> **kinematic variation**
- OR
- (use joint torque/position and load cell) -> sensor fusion, **extended kalman filter**
- complicates control problem
 - nonlinear coupling dynamics
 - orthogonal subspaces (hybrid control)

Hybrid control?

- Force Control (Admittance) along the axis of the fishing line to maintain perfect tension
- Rigid Position Control in the perpendicular axes
- Motivation: stiffness to reorient rod optimally, while allowing compliance on the force adaptive pulling axis

Algorithmic extensions

More robust state switching capability

Current Finite State machine uses force thresholding, which is very “chattery”

Solutions/Considerations:

- Hysteresis (anisotropic force thresholds for entering and exiting a state) & Debouncing (time based thresholding/conditioning) to limit false/rapid triggering
- Bumpless transfer
 - smoothing filters
 - transition states?
- Tension break detection (rate of force change) and **funnel synthesis (bound tracking error)**
 - Drops into a safe, heavily damped recovery ‘funnel’ in case of line break preventing the robotic arm from violently whipping backward and breaking motors
 - funnel synthesis concept also useful for control regain
- Admittance Control hardware implementation (gain scheduling)

Algorithmic extensions

Noise reduction

Probabilistic state estimation (HMM, Bayesian filter)

[important in RL, partial observability/limited info scenarios]

- Frequency and derivative of the force to distinguish between fake disturbances (boat rocking, snag on seaweed) or an actual fish

OR

Just use vision (unless there are partial observability concerns)

Algorithmic extensions (AI)

Adaptive neural network

Augmentation of control loop with an adaptive neural network feedforward term that learns the stochastic frequency of the fish's pulling behavior, effectively predicting the required q_d before the tension error grows too large.

Sim-to-real: Using the simulation as an AI training tool to optimize fish tensioning behavior.

- Direct RL: neural network
- Layered RL: controller parameter tweaking
- sim pipeline can also be used for adaptive control tuning without AI

Hardware Developments

Redundantly actuated serial joint

- Emulate rod compliance with active control capability?
- Macro-micro manipulator structure
 - **heavy base joint** handle low-frequency, high-amplitude positioning (like tracking boat's movement)
 - A **smaller, redundantly actuated serial joint** near the tip can be tuned with a much higher control loop frequency to handle high-frequency stochastic disturbances
 - **inertial decoupling**: torque efficiency, jerky movements doesn't cause heavy base motor to respond
- Null space optimization
 - infinite joint configs.
 - initiate secondary optimization objectives (torque minimization)
- Dynamic impedance shaping
 - redundant motor acts as dynamically tunable physical stiffness unit that adheres to control needs

Hardware Developments

A **compliant segment** added to the end of the rod

- tune stiffness characteristics
- **mechanical LPF**, dealing with incredibly jerky fish motion in true fishing scenarios
- **bumpless transfer stabilization** (state transitions), reduce risky torque spikes

Hardware testing rig

- Actuatable object to mimic a fish that has planarized movement capabilities to perform varied experiments on our hardware system and its accompanying controller design

Practical Additions

Bringing the product to 'market'

- Tensioning reel mechanism
- actuated hooking system

Thank You

Sources

<https://www.researchgate.net/publication/344142164> Haptic Display Using Fishing Rod